

HARDIK RAUT

Innovator

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EXPERIENCE

Mechanical Engineering Intern Jan 2026 - Jun 2026

Dhruva Space Pvt Ltd. | Hyderabad

- Designed and optimized aerospace structures in CAD for high stiffness-to-mass ratio under extreme launch environments.
- Performed modal and structural FEA to evaluate natural frequencies, stiffness, and dynamic response using Altair Inspire.
- Executed Random Vibration Analysis (RVA) using launch PSD profiles, deriving RMS/ 3σ stresses and Factor of Safety for structural qualification.
- Engineered lightweight truss and topology-optimized structures incorporating DfAM constraints for metal additive manufacturing.
- Presented research work at the Simcenter Technology Conference 2026 in Pune, showcasing computational approaches to aerospace structural analysis and optimization.

Lead for Structures Subsystem May 2025 - May 2026

COEP Satellite Initiative | COEP Technological University, Pune

- Led cross-subsystem engineering collaboration to resolve structural, mechanical, electrical, and interface constraints throughout the satellite development cycle.
- Led ideation, design, and validation of structural components and mechanisms, translating mission requirements into manufacturable solutions.
- Built and mentored the Structures team, coordinating technical responsibilities and establishing structured design and verification workflows.
- Represented CSAT at the GeoSmart India Conference and participated in the India-Japan Space & Geospatial Roundtable, engaging with industry and international stakeholders.

Subsystem Member Jan 2023 - May 2025

COEP Satellite Initiative | COEP Technological University, Pune

- Developed proficiency in orbital thermal simulations to evaluate spacecraft structural thermal behavior across representative orbital conditions.
- Optimized CAD designs with focus on structural performance, weight reduction, and manufacturability.
- Studied vibrational behavior of flexible deployable solar arrays, analyzing structural dynamics and vibration response during deployment and operation.

SUMMARY

Hands-on mechanical engineer with a strong foundation in CAD design, structural simulation, and mechanism development, focused on transforming ideas into practical, manufacturable products.

Combines engineering design with a strong understanding of manufacturing processes and DFM principles, ensuring that performance and real-world producibility are considered together.

Brings an interdisciplinary approach driven by interests in electronics, programming, embedded systems, and prototyping, building solutions at the intersection of mechanical, electronic, and computational systems.

EDUCATION

COEP Technological University

B. Tech in Manufacturing Science and Engineering

Nov 2022 - Aug 2026

SKILLS

- CAD Design & 3D Modeling
- Product Design & New Product Development
- Mechanism Design & Development
- Prototyping & Iterative Development
- Design for Manufacturing (DFM)
- Design for Assembly (DFA)
- Engineering Drawings & GD&T

SOFTWARE

- SolidWorks
- Onshape
- Siemens NX
- Autodesk Inventor
- Altair Inspire
- Altair SimSolid
- Fusion
- Ansys Workbench

PROJECTS

Structural Optimization of a Rigid Boom Structure for a Patch Antenna for Satellite Applications

Internship Project at Dhruva Space for Final Semester Evaluation

Performed structural optimization of a satellite rigid boom structure to improve stiffness and dynamic performance, using CAD-based design and FEA to evaluate structural behavior under space-environment loading. Iterated the design considering structural integrity, mass, and manufacturability for practical fabrication. Presented the research paper for this project at the Simcenter Technology Conference.

Freewheel based Deployer for Satellite

Personal Project

Designed and prototyped a compact pull-type retractable boom deployer for CubeSat applications, featuring a dual-spindle freewheel mechanism and tension-driven ribbon actuation for controlled, reversible deployment. Developed the mechanism for smooth operation, passive overspeed protection, and compact sub-1U integration, with the architecture covered under a provisional patent application.

Water Based Propulsion System

Personal Project

Developed a compact satellite propulsion concept using onboard water electrolysis to generate propellant in orbit, integrating a vacuum-compatible electrolyzer with a micropropulsion thruster. Built functional 3D-printed prototypes and developed the system architecture toward space-grade implementation. Currently under design patent filing procedure.

Bionic Arm

Project under COEP Technological University

Developed an EMG sensor-actuated bionic arm as part of the academic requirement for the mini project, enabling the prosthetic hand to replicate user hand movements. Won 1st place in the UG Project Competition at COEP under the Best Social Impact category for the project.

PATENTS

A Deployable and Retractable Membrane Deployment System

Application Number – 202621073997 Filed on – Jun 15, 2026
Patent under COEP Satellite Initiative

Freewheel based Deployer for Satellite

Application Number – 202521120359 Filed on – Dec 2, 2025
Personal Patent

Electrolytic Thruster

Application Number – 473721-001 Filed on – Sep 16, 2025
Personal Patent

The Cloud Chamber

Application Number – 202421059911 Filed on – Aug 8, 2024
Personal Patent

AWARDS

UG Project Competition – Jun 21, 2025

Best Social Impact Category
Project Title: The Bionic Arm

Industrial Psychology – Dec 5, 2024

Best Paper Presenter Team
Project Title: Effect of Constructive Divergence on Problem Solving Approaches

CONFERENCES

Simcenter Technology Conference
Pune, 7th May 2026

GeoSmart India Conference
Delhi, 2nd – 5th Dec 2025

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